

DATA QUALITY AND DATA LOADING ASSIGNMENT

Dataset:

Order ID	Customer ID	Sales Amount	Order Date
O101	C001	4500	12-01-2024
O102	C002		Null
O103	C003	3200	O101
C001	4500	15-01-2024	
O104	C004	Three Thousand	2024/01/18
O105	C005	5100	12-01-2024

Q1) Data Understanding

Identify all data quality issues present in the dataset that can cause problems during data loading.

Answer:

1. Missing Values (Nulls)

- Order_Date contains a Null value.
- Missing values can break date parsing or downstream calculations.

2. Inconsistent Data Types

- Sales_Amount includes both numeric and text values:
 - 4500, 3200, 5100
 - "Three Thousand"
- This causes type mismatch errors in databases.

3. Inconsistent Date Formats

- Order_Date has multiple formats:
 - DD-MM-YYYY (e.g., 12-01-2024)
 - YYYY/MM/DD (e.g., 2024/01/18)

4. Duplicate Records

- Order_ID O101 appears twice, violating uniqueness.

5. Misaligned / Corrupted Rows

- Some values appear in incorrect columns due to structure issues.

6. Inconsistent Record Lengths

- Some rows do not follow consistent column structure.

7. Data Integrity Issues

- Values are placed in incorrect fields in some cases.

8. Lack of Clear Structure

- Column alignment is inconsistent, making mapping unclear.
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Q2) Primary Key Validation

Assume Order_ID is the Primary Key.

a) Is the dataset violating the Primary Key rule?

Yes, the dataset violates the Primary Key rule.

b) Which record(s) cause this violation?

- Order_ID **O101** appears **twice**, violating uniqueness.
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Q3) Missing Value Analysis

Which column(s) contain missing values?

a) Affected Record(s):

- Order_ID: O102
- Customer_ID: C002
- Sales_Amount: 4500
- Order_Date: NULL

b) Why is this risky?

- Breaks data integrity
 - Affects reporting and analytics
 - May violate NOT NULL constraints
 - Can cause ETL and BI failures
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Q4) Data Type Validation

Identify records where Sales_Amount violates expected data type rules.

a) Which record(s) will fail numeric validation?

- Order_ID: O105
- Sales_Amount: "Three Thousand"

b) What happens in SQL (DECIMAL column)?

- Load failure due to conversion error (strict mode)
- Row rejection in ETL process

- Possible NULL conversion in lenient mode
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Q5) Date Format Consistency

The `Order_Date` column has multiple formats.

a) List all date formats present:

- DD-MM-YYYY → 12-01-2024, 15-01-2024
- YYYY/MM/DD → 2024/01/18
- NULL (missing value)

b) Why is this a problem?

- Parsing errors during loading
 - Incorrect date interpretation
 - Load failures in strict systems
 - Incorrect time-based analysis
 - ETL pipeline failures
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Q6) Load Readiness Decision

a) Should this dataset be loaded directly?

No.

b) Justification:

1. Duplicate `Order_ID` (O101) violates primary key rule
 2. Invalid `Sales_Amount` ("Three Thousand") causes type mismatch
 3. Inconsistent date formats and missing values
 4. Structural and alignment issues in dataset
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Q7) Pre-Load Validation Checklist

- Schema validation (correct columns and structure)
- Primary key uniqueness check
- Data type validation (numeric and date formats)
- Missing value detection
- Duplicate record detection
- Date format standardization

- Range and domain validation
 - Data consistency checks
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Q8) Cleaning Strategy

1. Fix structural alignment of dataset
 2. Remove or correct duplicate Order_ID (O101)
 3. Convert Sales_Amount to numeric (fix "Three Thousand")
 4. Standardize Order_Date to YYYY-MM-DD format
 5. Handle missing Order_Date values
 6. Validate relationships between fields
 7. Perform final validation before loading
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Q9) Loading Strategy

a) Should Full Load or Incremental Load be used?

Full Load (after cleaning)

b) Justification:

- Dataset contains multiple data quality issues
 - No reliable baseline for incremental updates
 - Full load ensures clean and consistent dataset
 - Prevents propagation of errors
 - Ensures accurate reporting
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Q10) BI Impact Scenario

a) Incorrect results in Total Sales KPI:

- Inflated sales due to duplicate Order_ID (O101)
- Missing or incorrect values due to invalid data
- Incomplete aggregation due to NULL values

b) Records causing misleading insights:

- O101 → duplicate counting
- O105 → invalid Sales_Amount
- O102 → NULL Order_Date

- Misaligned rows → incorrect mapping

c) Why BI tools don't detect issues:

- Assume data is already cleaned
 - Do not enforce strict validation rules
 - Convert invalid values to NULL silently
 - Focus on visualization, not data quality checks
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